

Data 211 Project

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```
# Load library
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
library(readxl)
```

```
## Warning: package 'readxl' was built under R version 4.5.3
```

```
#data
sleep_data <- read_excel("sleep_data.xlsx", range = "A1:B22")
```

Summary Statistics

```
summary(sleep_data$sleep_hours)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      3.500   6.000   7.000   6.881   8.000  10.500
```

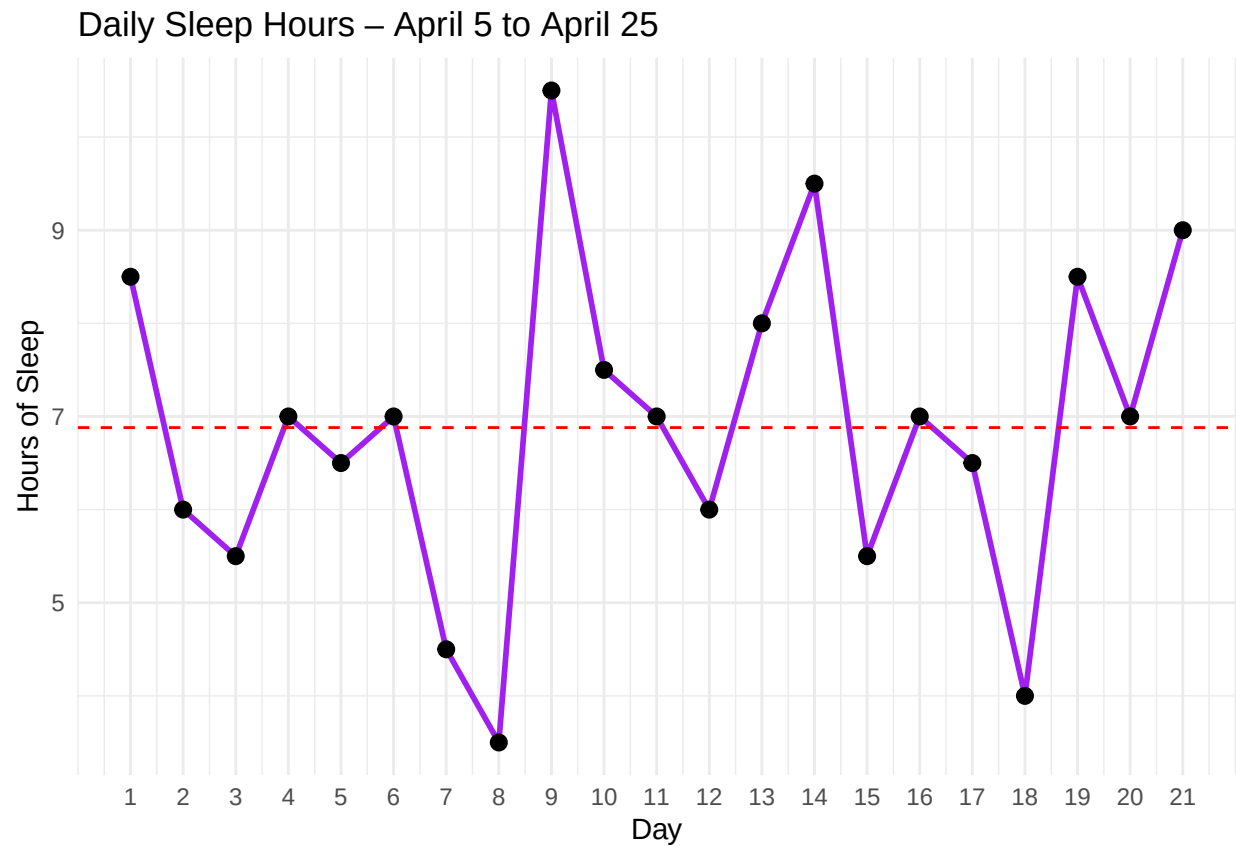
```
sd(sleep_data$sleep_hours)
```

```
## [1] 1.767093
```

Visualization

```
# Line Graph
ggplot(sleep_data, aes(x = day, y = sleep_hours)) +
  geom_line(color = "purple", size = 1) +
  geom_point(color = "black", size = 2.5) +
  geom_hline(yintercept = mean(sleep_data$sleep_hours),
             linetype = "dashed", color = "red") +
  labs(
    title = "Daily Sleep Hours - April 5 to April 25",
    x = "Day",
    y = "Hours of Sleep"
  ) +
  scale_x_continuous(breaks = 1:21) +
  theme_minimal()
```

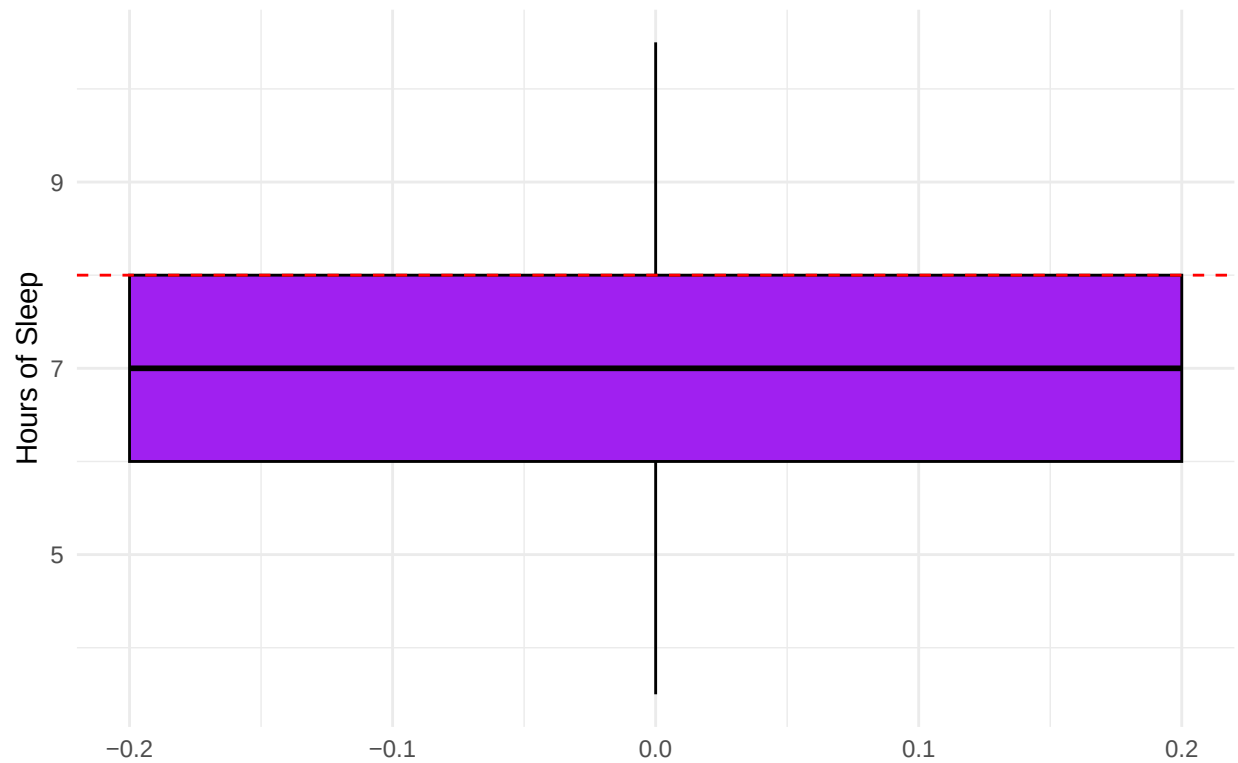
```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



The line graph shows my nightly sleep hours fluctuating throughout the 21-day period. There is no consistent pattern. Some nights I slept well above average while others dropped significantly. Most nights fall between 6 and 8 hours, with day 9 being the highest at 10.5 hours and day 8 being the lowest at 3.5 hours

```
#BoxPlot
ggplot(sleep_data, aes(y = sleep_hours)) +
  geom_boxplot(fill = "purple", color = "black", width = 0.4) +
  geom_hline(yintercept = 8, linetype = "dashed", color = "red") +
  labs(
    title = "Sleep Hours: Spread and Variability",
    x = " ",
    y = "Hours of Sleep"
  ) +
  theme_minimal()
```

Sleep Hours: Spread and Variability

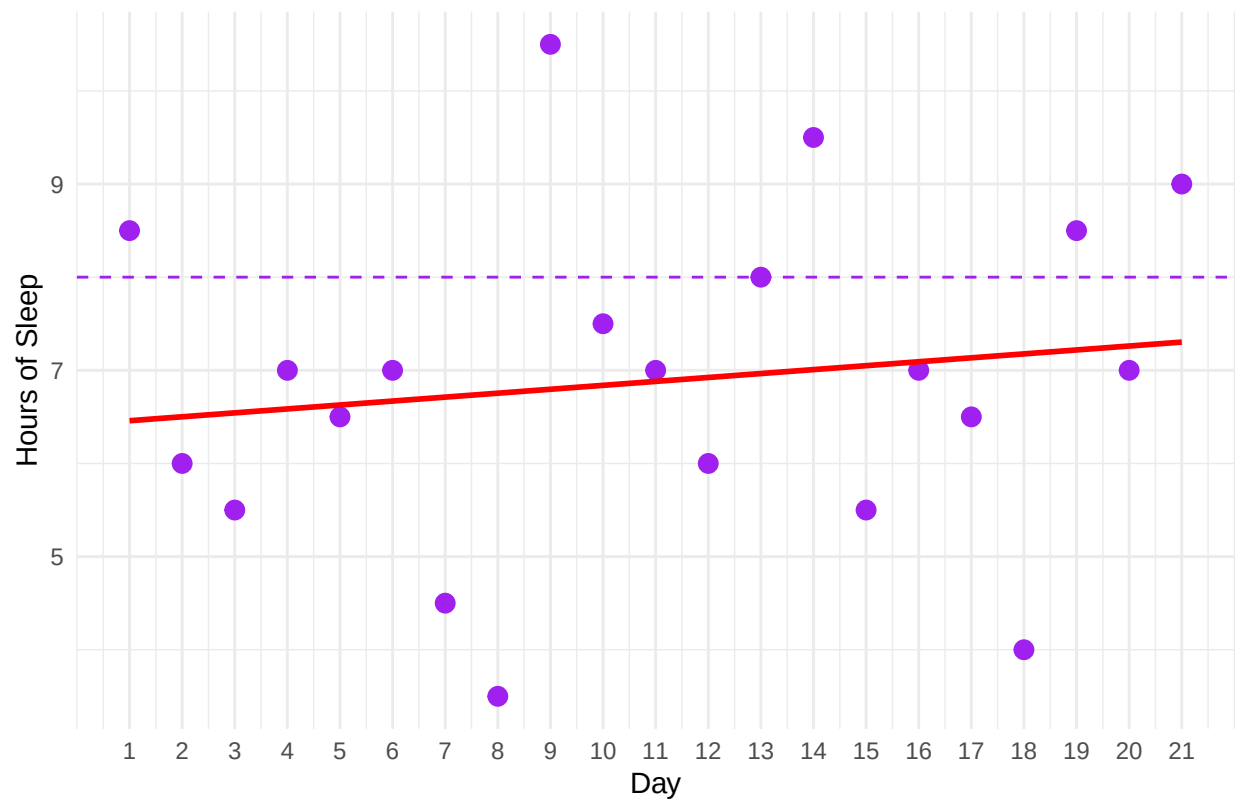


The box plot shows that the median sleep is 7.0 hours and the middle 50% of nights fall roughly between 5.5 and 8.0 hours. The dashed line at 8 hours shows that my median sits below the recommended amount. The spread of the box indicates moderate variability in my sleep habits.

```
#Scatterplot
ggplot(sleep_data, aes(x = day, y = sleep_hours)) +
  geom_point(color = "purple", size = 3) +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
  geom_hline(yintercept = 8, linetype = "dashed", color = "purple") +
  labs(
    title = "Sleep Trend Over 21 Days",
    x = "Day",
    y = "Hours of Sleep"
  ) +
  scale_x_continuous(breaks = 1:21) +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Sleep Trend Over 21 Days



The trend line shows a slight upward slope, suggesting my sleep hours marginally increased over the 21-day period. However, the high variability among the data points indicates this trend is weak and not conclusive.

Statistical Test

```
t.test(sleep_data$sleep_hours, mu = 8, alternative = "two.sided")
```

```
##
## One Sample t-test
##
## data:  sleep_data$sleep_hours
## t = -2.902, df = 20, p-value = 0.008814
## alternative hypothesis: true mean is not equal to 8
## 95 percent confidence interval:
##  6.076581 7.685324
## sample estimates:
## mean of x
##  6.880952
```

Interpretation

The one-sample t-test produced a t-statistic of -2.902 and a p-value of 0.0088, which is below the significance level of 0.05. I reject the null hypothesis. There is strong statistical evidence that my average nightly sleep of 6.88 hours is significantly less than the recommended 8 hours.